

# İBRAHİM ETHEM DEVECİ

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I work on reasoning in large language models, specifically on whether and how they can be made to reason better, not just perform better on benchmarks. My approach draws on interactionist theories of reasoning from cognitive science and connects them to reinforcement learning. The broader question underneath all of it is what it would even mean for a computational model to reason.

## EDUCATION

### METU, Graduate School of Informatics, Ankara, Turkey

Sep. 2024 – Present

- Doctoral Student in Cognitive Science

### METU, Graduate School of Informatics, Ankara, Turkey

Oct. 2020 – May 2024

- Master of Science in Cognitive Science
- GPA: 4.00
- Thesis: “Transformer Models for Translating Natural Language Sentences into Formal Logical Expressions”

### Koç University, College of Engineering, İstanbul, Turkey

Sep. 2014 – June 2020

- Bachelor of Science in Computer Engineering
- Track Program Certificate in Brain and Cognitive Sciences

## RESEARCH EXPERIENCE

### UNI-LM: A Universal Framework for Language Modeling

(BİDEB-2232 / Lider Araştırmacılar Programı), Ankara, Turkey

June 2025 – Present

- Investigating the intersection of cognitive science and AI by integrating interactionist theories of reasoning into multi-agent systems.
- Designing and implementing RL-based training pipelines and dense reward functions to enhance multi-agent reasoning, faithfulness, and generalization.

### Spatial Audio Signal Processing for 6-DoF Immersive Environments

(TÜBİTAK 1001 Project), Ankara, Turkey

Mar. 2022 – July 2023

- Worked as a TÜBİTAK (The Scientific and Technological Research Council of Türkiye) graduate scholar under Hüseyin Hacıhabiboğlu, PhD, developing virtual reality environments and 3D object models using Reality Capture for spatial sound processing experiments and algorithm testing in Unity.

### Koç University Research in Applied Memory (KURAM), İstanbul, Turkey

Jan. 2018 – June 2020

- Worked as an undergraduate research assistant in KURAM LAB under Sami Gülgöz, PhD, assisting in designing and running performance tasks using PsychoPy, collecting and analyzing behavioral data with Qualtrics, Excel, and SPSS, receiving training in experimental design, programming, and data analysis, and participating in weekly research meetings.

## SCHOOLS, WORKSHOPS, and TRAINING

### 36th European Summer School in Logic, Language, and Information (ESSLLI 2025)

July – Aug. 2025

Ruhr University Bochum, Bochum, Germany

- Awarded the RUB-Stiftung Fellowship. Participated in interdisciplinary coursework spanning logic, formal semantics, argumentation theory, and neural networks.

### 15th Lisbon Machine Learning Summer School (LxMLS 2025)

July 2025

Instituto Superior Técnico, Lisbon, Portugal

- Participated in an intensive program covering neural networks, transformers, sequence models, causality, and vision-language models, including hands-on lab sessions.

### Artificial Intelligence: Fundamental Models, Problems, and Future

July 2024

TED University, Ankara, Turkey

- Attended lectures on large language models, hybrid modeling approaches, transformers, and the history of NLP.

- Completed specialized coursework in philosophy of mind, learning and cognition, logic, rationality, and artificial intelligence.

**Artificial Intelligence and Formal Ontology Workshop**

Sept. 2019

İstanbul University, İstanbul, Turkey

- Engaged with topics including ontology design principles, data integration, OWL, and the semantic web.

**Fundamental and Advanced Levels Logic Summer School**

Oct. 2018

Boğaziçi University, İstanbul, Turkey

- Studied classical and non-classical logics, inference types, computability, metalogic, and neural networks.

**PUBLICATIONS AND PRESENTATIONS****PUBLICATIONS**

- Olgun, O., Deveci, İ. E., & Hacıhabiboğlu, H. (2026). Parametric sound field interpolation for scene-based navigable immersive audio. *IEEE Transactions on Audio, Speech and Language Processing*, 34, 1608–1620. <https://doi.org/10.1109/TASLPRO.2026.3671646>
- Deveci, İ. E. (2025). Yapay zekada sembolik ve sembolik olmayan ayrımı üzerine [On the distinction between symbolic and subsymbolic in artificial intelligence]. In Z. Göz (Ed.), *Yapay zekâ felsefesi* (pp. 173–236). Doruk Yayınları.
- Deveci, İ. E., & Ataman, D. (2025). The Ouroboros of benchmarking: Reasoning evaluation in an era of saturation. *NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle: Benchmarks, Emergent Abilities, and Scaling*. <https://arxiv.org/abs/2511.01365>
- Chang, T. A., Arnett, C., Eldesokey, A., ... Deveci, İ. E., ... Wang, J. (2025). Global PIQA: Evaluating physical commonsense reasoning across 100+ languages and cultures [Preprint]. *arXiv*. <https://arxiv.org/abs/2510.24081>
- Deveci, İ. E. (2025). Transformer models for translating natural language sentences into formal logical expressions. In V. Gradimondo & E. E. Rosina (Eds.), *Proceedings of the ESSLLI 2025 Student Session, 36th European Summer School in Logic, Language, and Information* (pp. 162–171). <https://philarchive.org/rec/GRAPOT-20>
- Deveci, İ. E., Küskü, M., & Özkurt, T. E. (2021). Makinenin beni ve ötekisi [Machine's self and other]. *Pasajlar Sosyal Bilimler Dergisi*, 9 (Yapay Zekâ)

**CONFERENCE PRESENTATIONS**

- Deveci, İ. E. (2025, July 1–3). Evaluating Compositionality in Large Language Models through Natural Language to First-Order Logic Translation [Poster presentation]. *Joint IACAP/AISB Conference on Philosophy of Computing and AI (IACAP/AISB-25)*, University of Twente, Enschede, Netherlands.
- Deveci, İ. E. (2024, December 6). Why Machine Learning Needs Task-specific Metrics: A Case Study on Semantic Parsing Models [Poster presentation]. *METU Open Research Day*, Middle East Technical University, Ankara, Turkey.
- Deveci, İ. E. (2024, June 1). Transformer Models for Translating Natural Language Sentences into Formal Logical Expressions [Poster presentation]. *ISBCS 2024*, Middle East Technical University, Ankara, Turkey.
- Deveci, İ. E., Zambak, A. F., & Temürcü, C. (2023, December 13). Bridging Words and Symbols: Using Transformers for Translating Sentences into Logical Expressions [Poster presentation]. *METU Open Research Day*, Middle East Technical University, Ankara, Turkey.
- Deveci, İ. E., & Başoğlu, C. (2023, May 7). The Main Problem with Neuroimaging Research in Explaining Cognitive Faculties [Poster presentation]. *ISBCS 2023*, Özyeğin University, İstanbul, Turkey.
- Şahin, Y., & Deveci, İ. E. (2023, May 7). Against Representations That Do Not Represent: Reply to Thomson and Piccinini [Poster presentation]. *ISBCS 2023*, Özyeğin University, İstanbul, Turkey.

## WORK EXPERIENCE

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- Coca Cola İçecek, Intern, Istanbul, Turkey** 2019 (July – August)
- Developed augmented reality applications using Unity, Arduino, and Raspberry Pi within the Digital Technologies (DT) Department.
- IAS, Intern, Istanbul, Turkey** 2017 (July - August)
- Built client-server applications in the IT Department using Java RMI and NSSM.
- Koç University, Work-Study Program, Dean of Students, Istanbul, Turkey** 2017 (March – June)
- Provided academic and logistical support for students with disabilities, including recording and digitizing lecture materials for the university library.

## LEADERSHIP and VOLUNTEERING EXPERIENCE

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- Founder and President of METU Cognitive Science Society** 2024 – 2025
- Organized and moderated introductory lectures and discussion groups on cognitive science.
- Coordinator in CogIST** 2021-2024
- Translated, edited, and reviewed cognitive science articles as editor-in-chief; coordinated both in-person and online events.
- Vice President of Koç University Philosophy Society** 2019-2020
- Organized and led bi-weekly discussion sessions and workshops on various philosophical topics.

## SKILLS AND INTERESTS

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- Technical Skills:** Python, PyTorch, Lisp, spaCy, Unity, LaTeX, MS Office, Adobe Photoshop
- Experience with:** R, Java, C#, C/C++, MATLAB, Reality Capture, SPSS, PsychoPy, Qualtrics
- Languages:** Turkish (native), English (fluent)
- Interests:** Argumentation theory and computational argumentation, artificial intelligence, cognitive science, large language models, logic, natural language processing, philosophy of artificial intelligence, mind, mathematics, language and action, Aikido and martial arts

## REFERENCES

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- Assist. Prof. Dr. Duygu Ataman**  
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Graduate School of Informatics, Middle East Technical University  
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*Relationship: Doctoral Advisor and Project Supervisor*
- Assoc. Prof. Dr. Aziz Fevzi Zambak**  
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